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Server device and filter replacement reminder method thereof

Abstract

The disclosure provides a server device and a filter replacement reminder method thereof. The filter replacement reminder method is suitable for the server device including a filter and includes the following steps. A current operation temperature of a target component is obtained by using a component temperature sensor. A working state of a heat dissipation component is detected. An ambient temperature is obtained by using an ambient temperature sensor. A reminder to replace the filter is provided by using a human-machine interface according to the current operation temperature, the working state of the heat dissipation component, and the ambient temperature.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 112101816, filed on Jan. 16, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure is related to a server device, and in particular, to a server device and a filter replacement reminder method thereof.

Description of Related Art

(3) With the advancement of technology, server devices providing various functions and services have been widely used in people's daily life. The server devices generally have powerful computing power and a large amount of storage space, so they may complete a large amount of work and store a large amount of file data in a short period of time, so as to provide the various services for a large number of users. Many components inside the server device will generate heat energy during operation, so the server device is generally provided with vents and fans to discharge the heat energy inside the server device. In order to prevent the dust in the air from entering the inside of the server device through the vent and affect the performance of the server device, the vent of the server device is generally provided with a filter. The filter may prevent the dust in the air from being brought into the inside of the server device. It is conceivable that as the working time of the server device increases, the dust will gradually accumulate on the filter. A clogged filter will result in a reduction in the air intake, thereby reducing the heat dissipation efficiency.

(4) Therefore, in order to maintain a smooth airflow through the vent, the filter with heavy dust accumulation must be replaced to avoid a decrease in heat dissipation performance and a bad impact on the performance of the server device. Currently, the replacement timing of the filter is based on a rule of thumb, and the server device will prompt the user to replace the filter according to a fixed time limit. For example, whenever the total working time of the server device reaches 6 months, the user is reminded to replace the filter. However, there are a variety of mechanical designs and heat dissipation designs of the server device, and the working state and environment of the server device are also different. Therefore, replacing the filter according to a fixed time limit is not a way to truly replace the filter according to the dust accumulation state. If the filter has serious dust accumulation and is not replaced in time, it will lead to poor operation performance of the server device, and even cause failure of the components in the server device.

SUMMARY

(5) The disclosure provides a filter replacement reminder method and a server device, which may allow the filter of the server device to be replaced at an appropriate time.

(6) An embodiment of the disclosure provides a filter replacement reminder method, which is suitable for a server device including a filter, and includes the following steps. A current operation temperature of a target component is obtained by using a component temperature sensor. A working state of a heat dissipation component is detected. An ambient temperature is obtained by using an ambient temperature sensor. A reminder to replace the filter is provided by using a human-machine interface according to the current operation temperature, the working state of the heat dissipation component, and the ambient temperature.

(7) An embodiment of the disclosure provides a server device, which includes a filter, a component temperature sensor, an ambient temperature sensor, a target component, a heat dissipation component, a human-machine interface, and a controller. The controller is coupled to the component temperature sensor, the ambient temperature sensor, the target component, the heat dissipation component, and the human-machine interface, and is used to execute the following steps. A current operation temperature of the target component is obtained by using the component temperature sensor. A working state of the heat dissipation component is detected. An ambient temperature is obtained by using the ambient temperature sensor. According to the current operation temperature, the working state of the heat dissipation component, and the ambient temperature, a reminder to replace the filter is provided by using the human-machine interface.

(8) Based on the above, in the embodiment of the disclosure, the component temperature sensor may sense the current operation temperature of the target component, and the current operation temperature of the target component may be used to determine the dust accumulation state of the filter. Thus, the reminder to replace the filter may be provided to the user according to the current operation temperature of the target component, the working state of the heat dissipation component, and the ambient temperature. Based on this, the user may know the appropriate time to

replace the filter, thus avoiding the late replacement of the filter and causing poor performance or failure of the server device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic block diagram of a server device according to an embodiment of the disclosure.

(2) FIG. **2** is a flow diagram of a filter replacement reminder method according to an embodiment of the disclosure.

(3) FIG. **3** is a flow diagram of a filter replacement reminder method according to an embodiment of the disclosure.

(4) FIG. **4**A is a schematic block diagram of a server device according to an embodiment of the disclosure.

(5) FIG. **4**B is a schematic diagram of a server device according to an embodiment of the disclosure.

(6) FIG. **5** is a flow diagram of a filter replacement reminder method according to an embodiment of the disclosure.

(7) FIG. **6** is a flow diagram of providing a maximum performance of a target component according to an embodiment of the disclosure.

(8) FIG. **7** is a flow diagram of providing a suggested service life of a filter according to an embodiment of the disclosure.

(9) FIG. **8** is a schematic diagram of estimating a service life of a filter according to an embodiment of the disclosure.

(10) FIG. **9** is a schematic diagram of estimating a service life of a filter according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(11) Parts of the embodiments of the disclosure will be described in details below with reference to the accompanying drawings. For the reference numerals used in the following description, when the same reference numerals appearing in different drawings will be regarded as the same or similar components. These embodiments are only a part of the disclosure, and do not disclose all possible implementation modes of the disclosure. Rather, these embodiments are only examples of the devices and methods within the scope of the disclosure.

(12) Referring to FIG. **1**, FIG. **1** is a schematic diagram of a server device according to an embodiment of the disclosure. A server device **100** may be, for example, a tower server, a rack server, a blade server, a multi-node server, or other types of servers, which is not limited in the disclosure. The server device **100** may include a filter F**1**, a component temperature sensor **110**, an ambient temperature sensor **120**, a target component **130**, a heat dissipation component **140**, a human-machine interface **150**, and a controller **160**.

(13) The filter F**1** may be disposed on the casing or the rack of the server device **100** to prevent external dust from entering the server device **100**. In some embodiments, the filter F**1** may be disposed at the air inlet. The dust accumulation state of the filter F**1** will affect the air intake flow, so the dust accumulation state of the filter F**1** will affect the heat dissipation efficiency of the server device **100**. The component temperature sensor **110** is used to sense the operation temperature of the target component **130**. The target component **130** may be a central processing unit (CPU), a hard disk, a memory module, or other electronic components in the server device **100** that generate heat during operation. The memory module is, for example, a dual in-line memory module (DIMM), but not limited thereto. In an embodiment, when the target component **130** is a CPU, the component temperature sensor **110** may be a built-in temperature sensor of the CPU. Alternatively,

in an embodiment, when the target component **130** is a memory module, the component temperature sensor **110** may be a built-in temperature sensor of the memory module.
(14) The ambient temperature sensor **120** is used to sense the ambient temperature of the environment where the server device **100** is located, and may be disposed at a position away from the electronic components that generate heat.
(15) The heat dissipation component **140** may include a fan, a cooling chip, a water-cooled heat dissipation device, other heat dissipation devices, or a combination of these devices. The heat dissipation component **140** provides a heat dissipation function, which may take away the heat energy inside the server device **100**.
(16) The human-machine interface **150** may include one or more input devices, such as a touch screen, a keyboard, a mouse, or buttons. The human-machine interface **150** may also include one or more output devices, such as a display, a speaker, or a lighting device. The user of the server device **100** may interact with the server device **100** through the human-machine interface **150**. In some embodiments, the controller **160** may provide a reminder to the user to replace the filter F**1** through the output device of the human-machine interface **150**.
(17) The controller **160** is coupled to the component temperature sensor **110**, the ambient temperature sensor **120**, the target component **130**, the heat dissipation component **140**, and the human-machine interface **150**. In some embodiments, the controller **160** may be a baseboard management controller (BMC) of the server device **100**.
(18) FIG. **2** is a flow diagram of a filter replacement reminder method according to an embodiment of the disclosure. Referring to FIG. **1** and FIG. **2**, the method of the embodiment is suitable for the server device **100** in the above-mentioned embodiment, and the detailed steps of the filter replacement reminder method of the embodiment will be described below in combination with the various components in the server device **100**.
(19) In step S**210**, the controller **160** obtains the current operation temperature of the target component **130** by using the component temperature sensor **110**. The component temperature sensor **110** may sense the current operation temperature of the target component **130** and provide the current operation temperature to the controller **160**. The operation temperature of the target component **130** is positively related to the ambient temperature. In some embodiments, the operation temperature of the target component **130** is also positively related to the load state of the target component **130**.
(20) In step S**220**, the controller **160** detects the working state of the heat dissipation component **140**. In some embodiments, the heat dissipation component **140** may be a fan, and the working state of the heat dissipation component **140** may be the fan speed. When the working state of the heat dissipation component **140** is abnormal, the operation temperature of the target component **130** will increase abnormally.
(21) In step S**230**, the controller **160** obtains the ambient temperature by using the ambient temperature sensor **120**. The ambient temperature sensor **120** may sense the ambient temperature and provide the ambient temperature to the controller **160**. The ambient temperature may affect the operation temperature of the target component **130**. An increase in the ambient temperature will increase the operation temperature of the target component **130**.
(22) In step S**240**, the controller **160** provides a reminder to replace the filter F**1** by using the human-machine interface **150** according to the current operation temperature, the working state of the heat dissipation component **140**, and the ambient temperature. In some embodiments, the controller **160** may control the luminous color, the flashing frequency, or whether the light is turned on or not of the lighting device, so as to notify the user whether the filter F**1** needs to be replaced. Alternatively, in some embodiments, the controller **160** may control the display to display a reminder message to notify the user whether the filter F**1** needs to be replaced.
(23) Specifically, as the use time increases, more and more dust will accumulate on the filter F**1**. The heat dissipation efficiency of the server device **100** will also decrease as the dust accumulated

on the filter F**1** increases, thereby affecting the operation temperature of the target component **130**. It may be known that the current operation temperature of the target component **130** may be used to evaluate the dust accumulation state of the filter F**1**. It should be noted that both the ambient temperature and the working state of the heat dissipation component **140** will also affect the current operation temperature of the target component **130**. Therefore, the controller **160** may determine whether to replace the filter F**1** according to the current operation temperature, the working state of the heat dissipation component **140**, and the ambient temperature, and decide whether to use the human-machine interface **150** to provide a reminder for replacing the filter F**1**. When the controller **160** determines that the filter F**1** needs to be replaced, the controller **160** controls the human-machine interface **150** to provide a reminder to replace the filter F**1**. In this way, the dust accumulation state of the filter F**1** is determined according to the current operation temperature of the target component **130**, and the user may be reminded to replace the filter F**1** at an appropriate time.

(24) In some embodiments, the controller **160** may determine whether the current operation temperature of the target component **130** is greater than or equal to a threshold value, so as to determine whether to provide a reminder to replace the filter F**1**. In some embodiments, the aforementioned threshold value may be a rated temperature of the target component **130**. In some embodiments, in response to the current operation temperature being greater than or equal to the rated temperature of the target component **130**, the controller **160** may provide a reminder to replace the filter F**1** according to the working state of the heat dissipation component **140** and the ambient temperature. In some embodiments, in response to the current operation temperature being greater than or equal to the rated temperature, the controller **160** may determine whether to provide a reminder to replace the filter F**1** according to whether the working state of the heat dissipation component **140** is normal or abnormal. In some embodiments, in response to the current operation temperature being greater than or equal to the rated temperature, the controller **160** may determine whether to provide a reminder to replace the filter F**1** according to whether the ambient temperature is too high. Alternatively, in some embodiments, the controller **160** may look up a table according to the ambient temperature, the working state of the heat dissipation component **140**, and the current operation temperature to determine whether to provide a reminder to replace the filter F**1**.

(25) FIG. **3** is a flow diagram of a filter replacement reminder method according to an embodiment of the disclosure. Referring to FIG. **1** and FIG. **3**, the method of the embodiment is suitable for the server device **100** in the above-mentioned embodiment, and the detailed steps of the filter replacement reminder method of the embodiment will be described below in combination with the various components in the server device **100**.

(26) In step S**310**, the controller **160** obtains the current operation temperature of the target component **130** by using the component temperature sensor **110**. It is worth mentioning that, in some embodiments, the controller **160** may first notify the target component **130** to operate at a fully loaded state (e.g., maximum utilization or maximum usage). After the target component **130** is operating at a fully loaded state, the controller **160** then obtains the current operation temperature of the target component **130** by using the component temperature sensor **110**. In step S**320**, the controller **160** determines whether the current operation temperature is greater than or equal to the rated temperature of the target component **130**. The rated temperature is the maximum operation temperature specified by the manufacturer of the target component **130** for the target component **130**. According to the heat dissipation design of the server device **100**, when the ambient temperature meets the specification and the heat dissipation component **140** works normally, the operation temperature of the target component **130** will not exceed the rated temperature of the target component **130**.

(27) If the determination in step S**320** is negative, it means that the filter F**1** does not need to be replaced yet, and so returns to step S**310**. If the determination in step S**320** is positive, it means that the filter F**1** may need to be replaced. Therefore, if the determination in step S**320** is positive, in

step S**330**, the controller **160** detects the working state of the heat dissipation component **140**. In step S**340**, the controller **160** determines whether the working state of the heat dissipation component **140** is normal. In an embodiment, the heat dissipation component **140** may be a fan. The controller **160** may detect the fan speed, and determine whether the fan speed is normal. For example, the controller **160** may control the fan to operate at the maximum speed through a control signal, and determine whether the actual speed reported by the fan is the maximum speed.
(28) If the determination in step S**340** is negative, it means that the working state of the heat dissipation component **140** is abnormal. Therefore, if the determination in step S**340** is negative, in step S**390**, the controller **160** provides a reminder to inspect the heat dissipation component **140** by using the human-machine interface **150**. On the other hand, if the determination in step S**340** is positive, it means that the working state of the heat dissipation component **140** is normal. Therefore, if the determination in step S**340** is positive, in step S**350**, the controller **160** obtains the ambient temperature by using the ambient temperature sensor **120**. In step S**360**, the controller **160** determines whether the ambient temperature is lower than an ambient temperature threshold. The ambient temperature threshold may be set according to actual applications, which is not limited in the disclosure.
(29) If the determination in step S**360** is negative, it means that the ambient temperature is too high. Therefore, if the determination in step S**360** is negative, in step S**380**, the controller **160** provides a reminder to reduce the ambient temperature by using the human-machine interface **150**. On the other hand, if the determination in step S**360** is positive, it means that the ambient temperature is not too high. Therefore, if the determination in step S**360** is positive, in step S**370**, the controller **160** provides a reminder to replace the filter F**1** by using human-machine interface **150**.
(30) Based on the flow diagram in FIG. **3**, when the current operation temperature is greater than or equal to the rated temperature of the target component, the controller **160** may provide a reminder to replace the filter F**1** in response to the working state of the heat dissipation component **140** being normal and the ambient temperature being lower than the ambient temperature threshold. In other words, under the condition that the working state of the heat dissipation component **140** is normal and the ambient temperature is lower than the ambient temperature threshold, if the current operation temperature is greater than or equal to the rated temperature of the target component, it means that the heat dissipation efficiency has been reduced due to the filer F**1** being blocked to a considerable extent, so replacement is necessary. Otherwise, when the current operation temperature is greater than or equal to the rated temperature of the target component, the controller **160** may determine not to provide a reminder to replace the filter F**1** because the working state of the heat dissipation component **140** is abnormal or the ambient temperature is not less than the ambient temperature threshold.
(31) It should be noted that the above-mentioned embodiments are described with one target component **130** as an example, but the disclosure does not limit the number of target components. In other embodiments, the controller **160** may determine whether the current operation temperatures of the multiple target components are greater than or equal to the corresponding rated temperatures. For example, the controller **160** may respectively determine whether the current operation temperature of the CPU is higher than the rated temperature of the CPU, and determine whether the current operation temperature of the memory module is higher than the rated temperature of the memory module. Therefore, the controller **160** may determine whether to provide a reminder to replace the filter F**1** according to the current operation temperatures of the multiple target components.
(32) FIG. **4**A is a schematic block diagram of a server device according to an embodiment of the disclosure. Referring to FIG. **4**A, a server device **400** may include the filter F**1**, the component temperature sensor **110**, the ambient temperature sensor **120**, the target component **130**, the heat dissipation component **140**, the human-machine interface **150**, the controller **160**, and an inlet temperature sensor **170**, another component temperature sensor **180**, and another target component

190.
(33) The functions and the coupling relationships of the filter F**1**, the component temperature sensor **110**, the ambient temperature sensor **120**, the target component **130**, the heat dissipation component **140**, the human-machine interface **150**, and the controller **160** are similar to the functions and the coupling relationships of the foregoing embodiments, and will not be repeated here. It should be noted that the server device **400** of the embodiment further includes the inlet temperature sensor **170**, another component temperature sensor **180**, and another target component **190**. The inlet temperature sensor **170**, another component temperature sensor **180**, and another target component **190** are coupled to the controller **160**.
(34) The component temperature sensor **180** is used to sense the operation temperature of the target component **190**. The component temperature sensor **180** may be a built-in temperature sensor of the target component **190**. The target component **190** and the target component **130** may be the same device or different devices. For example, the target component **190** and the target component **130** may be two CPUs. Alternatively, the target component **130** may be a CPU, and the target component **190** may be a memory module.
(35) The inlet temperature sensor **170** is used to sense the inlet temperature of the target component **130**. The inlet temperature may also be referred to as the fan inlet temperature. The inlet temperature sensor **170** may be disposed between the fan and the target component **130**.
(36) FIG. **4**B is a schematic diagram of a server device according to an embodiment of the disclosure. Referring to FIG. **4**B, in the embodiment, the heat dissipation component **140** of the server device **400** may include fans **140***a* and **140***b*. The fans **140***a* and **140***b*, the target component **130**, and the target component **190** may be disposed on a motherboard P**1**. The filter F**1** may be disposed on one side of the fans **140***a* and **140***b*, and may be disposed at the air inlet of the casing. The target component **130** and the target component **190** are disposed on the other side of the fans **140***a* and **140***b*. Driven by the fans **140***a* and **140***b*, the air enters the server device **100** from the air inlet through the filter F**1**, and reaches the location of the target component **130** and the target component **190** along an airflow direction D**1**.
(37) The inlet temperature sensor **170** may be disposed on the air channel between the fan **140***a* and the target component **130** to sense the inlet temperature of the target component **130**. The ambient temperature sensor **120** may be disposed at a position away from the target component **130** and the target component **190**. The component temperature sensors **110** and **180** may be built-in temperature sensors of the target component **130** and the target component **190**, respectively.
(38) FIG. **5** is a flow diagram of a filter replacement reminder method according to an embodiment of the disclosure. Referring to FIG. **4**A, FIG. **4**B, and FIG. **5**, the method of the embodiment is suitable for the server device **400** in the above-mentioned embodiment, and the detailed steps of the filter replacement reminder method of the embodiment will be described below in combination with the various components in the server device **400**.
(39) It should be noted that in the embodiment, the controller **160** may also estimate the maximum performance and the service life of the filter F**1** according to the current operation temperature of the target component **130**, and let the user know the maximum performance of the component **130** and the service life of the filter F**1** through the human-machine interface.
(40) In step S**510**, the controller **160** may obtain the current operation temperature of the target component **130** by using the component temperature sensor **110**. In addition, the controller **160** may also obtain the current operation temperature of the target component **190** by using the component temperature sensor **180**. Therefore, in the subsequent steps, the controller **160** may determine whether to provide a reminder to replace the filter F**1** according to the current operation temperatures of the target components **130** and **190**, respectively.
(41) In step S**520**, the controller **160** provide the maximum performance of the target component **130** by using the human-machine interface **150** according to the current operation temperature of the target component **130**. Specifically, as the filter F**1** accumulates more dust, the heat dissipation

efficiency gradually deteriorates due to the reduction of the air intake, and the maximum performance of the target component **130** also decreases due to the deterioration of the heat dissipation efficiency. Here, the controller **160** may estimate the maximum performance that the target component **130** may achieve under the current dust accumulation state of the filter F**1**. In more detail, the controller **160** may estimate the maximum performance currently supported by the target component **130** according to the current operation temperature and the current performance of the target component **130**. For example, the target component **130** may be a CPU, and the maximum performance of the target component **130** estimated by the controller **160** is the highest power wattage of the CPU.

(42) Referring to FIG. **6**, FIG. **6** is a flow diagram of providing a maximum performance of a target component according to an embodiment of the disclosure. Step S**520** may be implemented as steps S**521** to S**523**. In step S**521**, the controller **160** obtains the inlet temperature of the target component **130** by using the inlet temperature sensor **170**.

(43) In step S**522**, the controller **160** estimates the maximum performance of the target component **130** according to the inlet temperature, the rated temperature, the current operation temperature, and the current performance of the target component **130**. In some embodiments, the current performance of the target component **130** is proportional to the difference between the current operation temperature and the inlet temperature. Therefore, the controller **160** may estimate the maximum performance of the target component **130** based on the current performance of the target component **130** according to the ratio of the first difference between the rated temperature and the inlet temperature to the second difference between the current operation temperature and the inlet temperature. In some embodiments, the target component **130** may be a CPU, and the controller **160** may estimate the maximum performance of the target component **130** according to the following formula (1). Formula (1) may be used to predict the maximum power wattage of the CPU when the current operation temperature of the CPU reaches the rated temperature.

(44) $$\text{CPUPower}_{support} = \text{CPUPower}_{average} \times \frac{T_{spec} - T_{in}}{T_j - T_{in}} \quad \text{formula(1)}$$

CPU Power.sub.support is the maximum performance of the target component **130**. CPU Power.sub.average is the current performance of the target component **130**. T.sub.spec is the rated temperature of the target component **130**. Tin is the inlet temperature of the target component **130**, and T.sub.j is the current operation temperature of the target component **130**. It is worth mentioning that, in some embodiments, the current performance of the target component **130** may be an average value of multiple current detection performances.

(45) In step S**523**, the controller **160** provides the maximum performance of the target component **130** by using the human-machine interface **150**. As the accumulation of dust on the filter F**1** becomes severe, the maximum efficiency of the target component **130** estimated by the controller **160** will decrease. The user may know the maximum performance of the target component **130** through the human-machine interface **150**, so as to clearly know the performance status of the server device **100**. Therefore, the user may also determine whether to replace the filter F**1** at one's discretion according to the maximum performance of the target component **130**.

(46) Returning to FIG. **5**, in step S**530**, the controller **160** provides a suggested service life of the filter F**1** by using the human-machine interface **150** according to the current operation temperature. Specifically, as the filter F**1** accumulates more dust, the heat dissipation efficiency gradually deteriorates due to the reduction in the air intake. In some embodiments, the controller **160** may regularly control the target component **130** to operate in a preset load state, and regularly measure multiple operation temperatures when the target component **130** operates in the preset load state. The operation temperatures operating at a preset load state will increase with time due to the gradual deterioration of the heat dissipation efficiency. Therefore, the controller **160** may estimate the service life of the filter F**1** according to the linear relationship of the operation temperature (i.e., the current operation temperature and the previous operation temperature) sensed at different time

points and the time.
(47) Referring to FIG. **7**, FIG. **7** is a flow diagram of providing a suggested service life of a filter according to an embodiment of the disclosure. In step S**531**, the controller **160** obtains the previous operation temperature sensed at the previous sensing time. That is, the controller **160** obtains the previous operation temperature of the target component **130** by using the component temperature sensor **110** at the previous sensing time. For example, the controller **160** may regularly obtain the operation temperature of the target component **130** every other day, a week, or a month. The current operation temperature is sensed at the current sensing time. It should be noted that both the previous operation temperature and the current operation temperature correspond to the preset load state of the target component **130**. The preset load state is, for example, a fully loaded state. That is, the controller **160** obtains the previous operation temperature and the current operation temperature through the component temperature sensor **110** when the target component **130** is operating under a preset load state.
(48) In step S**532**, the controller **160** estimates the service life of the filter F**1** according to the rated temperature of the target component **130** based on the linear relationship of the current operation temperature and the previous operation temperature relative to the previous sensing time and the current sensing time. For details, please refer to FIG. **8**, which is a schematic diagram of estimating a service life of a filter according to an embodiment of the disclosure. t.sub.n is the current sensing time. t.sub.n-1 is the previous sensing time. t.sub.x is the predicted time when the operation temperature of the target component **130** reaches the rated temperature T.sub.spec. T.sub.n is the current operation temperature. T.sub.n-1 is the previous operation temperature. The controller **160** may estimate the service life TL**1** of the filter F**1** according to the following formula (2).
(49) $\mathrm{TL1} = t_x - t_n = \frac{(T_{\mathrm{spec}} - T_n) \times (t_n - t_{n-1})}{T_n - T_{n-1}}$ formula(2)
(50) In step S**533**, the controller **160** provides a suggested service life of the filter F**1** by using the human-machine interface **150** according to the service life of the filter F**1**. Accordingly, the user may know the suggested service life of the filter F**1** through the human-machine interface **150**. Therefore, the user may also determine whether to replace the filter F**1** at one's discretion according to the suggested service life of the filter F**1**. Alternatively, the user may prepare for the replacement of the filter F1 in advance according to the suggested service life of the filter F**1**.
(51) Take the target component **130** is CPU for example, the rated temperature T.sub.spec of CUP is 100° ° C. In some embodiments, the suggested service life of the filter F**1** may be the service life TL**1** calculated according to the latest current operation temperature shown in FIG. **8**. Alternatively, in some embodiments, since the controller **160** may regularly calculate the service life of the filter F**1**, the controller **160** may use the minimum of the service life calculated at different time points as the suggested service life of the filter F**1**.
(52) In some embodiments, the controller **160** may also obtain another previous operation temperature sensed at another previous sensing time. It should be noted that both the previous operation temperature and another previous operation temperature correspond to the preset load state of the target component **130**. The preset load state is, for example, a fully loaded state. That is, the controller **160** obtains the previous operation temperature and another previous operation temperature through the component temperature sensor **110** when the target component **130** operates under a preset load state. Afterwards, the controller **160** may estimate another service lift of the filter F**1** according to the rated temperature of the target component **130** based on another linear relationship of the previous operation temperature and another previous operation temperature relative to the previous sensing time and another previous sensing time. The suggested service life may be the shorter of the service life and another service life.
(53) For details, please refer to FIG. **9**, which is a schematic diagram of estimating a service life of a filter according to an embodiment of the disclosure. t.sub.n is the current sensing time. t.sub.n-1 is the previous sensing time. t.sub.n-2 is another previous sensing time. t.sub.x and ty are the

predicted times when the operation temperature of the target component **130** reaches the rated temperature T.sub.spec. T.sub.n is the current operation temperature. T.sub.n-1 is the previous operation temperature. T.sub.n-2 is another previous operation temperature. The controller **160** may estimate the service life TL**1** of the filter F**1** according to formula (2). Moreover, the controller **160** may estimate another service life TL**2** of the filter F**1** according to the following formula (3).

(54) $$TL2 = t_y - t_{n-1} = \frac{(Tspec - T_{n-1}) \times (t_{n-1} - t_{n-2})}{T_{n-1} - T_{n-2}} \quad \text{formula(3)}$$

(55) In the example of FIG. **9**, take the target component **130** is CPU for example, the rated temperature T.sub.spec of CUP is 100° C. Therefore, the controller **160** may select the shorter of another service life TL**2** and the service life TL**1** as the recommended service life of the filter F**1**. The controller **160** may determine that the suggested service life may be another service life TL**2** due to another service life TL**2** being less than the service life TL**1**.

(56) In addition, in some embodiments, based on the same principle and operation process, the controller **160** may also estimate another service life of the filter F**1** according to the current operation temperature of another target component **190**. Thus, the suggested service life may be the shorter of the service life corresponding to the target component **130** and another service life corresponding to the target component **190**. In more detail, the controller **160** may obtain another current operation temperature and another previous operation temperature of another target component **190** by using another component temperature sensor **180**. The controller **160** may estimate another service life of the filter F**1** according to another rated temperature of another target component **190**, another current operation temperature, and another previous operation temperature. Therefore, the controller **160** may select the shorter of another service life corresponding to another target component **190** and the service life corresponding to the target component **130** as the suggested service life of the filter F**1**.

(57) Afterwards, returning to FIG. **5** again, at step S**540**, the controller **160** detects the working state of the heat dissipation component **140**. In step S**550**, the controller **160** obtains the ambient temperature by using the ambient temperature sensor **120**. In step S**560**, the controller **160** provides a reminder to replace the filter F**1** by using the human-machine interface **150** according to the current operation temperature of the target component **130**, the working state of the heat dissipation component **140**, and the ambient temperature. Similarly, the controller **160** may also provide a reminder to replace the filter F**1** by using the human-machine interface **150** according to the current operation temperature of the target component **190**, the working state of the heat dissipation component **140**, and the ambient temperature. The detailed implementation steps of step S**540** to step S**560** have been described in the foregoing embodiments, and will not be repeated here.

(58) It is worth mentioning that, in some embodiments, the user may set the inspection frequency of the filter F**1** through the software interface, such as every half a month or a week, which is not limited in the disclosure. The controller **160** may regularly execute the process shown in FIG. **5** according to the inspection frequency set by the user. Whenever the process shown in FIG. **5** is to be executed, the controller **160** may set the fan operation to the maximum speed and control the target component **130** to operate at a preset load state, so as to detect whether the fan speed is normal and to estimate the service life of the filter F**1**.

(59) To sum up, in the embodiment of the disclosure, the component temperature sensor may sense the current operation temperature of the target component, and the current operation temperature of the target component may be used to determine the dust accumulation state of the filter. Therefore, a reminder to replace the filter may be provided to the user according to the current operation temperature of the target component, the working state of the heat dissipation component, and the ambient temperature. Since the filter of the server device may be replaced at the appropriate time, it may not only avoid unnecessary waste caused by replacing the filter too early, but also avoid the noise and the power consumption of the continuous high-speed operation of the fan when the filter is replaced too late, the poor performance of the server device, or the failure of the components

within the server device.

(60) In addition, the suggested service life of the filter and the maximum performance of the target component may also be estimated according to the current operation temperature of the target component and be provided to the user through the human-machine interface. In this way, the user may clearly understand the dust accumulation status of the filter through the suggested service life of the filter and the maximum performance of the target component, and may determine whether to replace the filter at one's discretion or prepare a spare filter in advance. In addition, the filter replacement reminder method of the embodiment of the disclosure is suitable for any type of server device.

Claims

1. A filter replacement reminder method, suitable for a server device comprising a filter, comprising: obtaining a current operation temperature of a target component by using a component temperature sensor; detecting a working state of a heat dissipation component; obtaining an ambient temperature by using an ambient temperature sensor; and providing a reminder to replace the filter by using a human-machine interface by the current operation temperature, the working state of the heat dissipation component, and the ambient temperature, wherein the step of providing the reminder to replace the filter by using the human-machine interface by the current operation temperature, the working state of the heat dissipation component, and the ambient temperature comprises: providing the reminder to replace the filter by the working state of the heat dissipation component and the ambient temperature, in response to the current operation temperature being greater than or equal to a rated temperature of the target component.
2. The filter replacement reminder method according to claim 1, wherein the step of providing the reminder to replace the filter by the working state of the heat dissipation component and the ambient temperature in response to the current operation temperature being greater than or equal to the rated temperature of the target component comprises: providing the reminder to replace the filter, in response to the working state of the heat dissipation component being normal and the ambient temperature being lower than an ambient temperature threshold.
3. The filter replacement reminder method according to claim 1, wherein the heat dissipation component comprises a fan, and the working state of the heat dissipation component comprises a fan speed.
4. The filter replacement reminder method according to claim 1, wherein the target component comprises a central processing unit, a hard disk, or a memory module.
5. The filter replacement reminder method according to claim 1, further comprising: obtaining a previous operation temperature sensed at a previous sensing time, wherein the current operation temperature is sensed at a current sensing time; estimating a service life of the filter by a rated temperature of the target component based on a linear relationship of the current operation temperature and the previous operation temperature relative to the previous sensing time and the current sensing time; and providing a suggested service life of the filter by using the human-machine interface by the service life of the filter, wherein the previous operation temperature and the current operation temperature both correspond to a preset load state of the target component.
6. The filter replacement reminder method according to claim 5, further comprising: estimating an another service life of the filter by the rated temperature of the target component based on an another linear relationship of the previous operation temperature and an another previous operation temperature relative to the previous sensing time and an another previous sensing time, wherein the suggested service life is the shorter of the service life and the another service life, and the previous operation temperature and the another previous operation temperature both correspond to the preset load state of the target component.
7. The filter replacement reminder method according to claim 5, further comprising: obtaining an

another current operation temperature and the another previous operation temperature of an another target component by using an another component temperature sensor; and estimating the another service life of the filter by an another rated temperature of the another target component, the another current operation temperature, and the another previous operation temperature, wherein the suggested service life is the shorter of the service life and the another service life.

8. The filter replacement reminder method according to claim 1, wherein the target component comprises a central processing unit, and the method further comprises: obtaining an inlet temperature of the target component by using an inlet temperature sensor; estimating a maximum performance of the target component by the inlet temperature, a rated temperature, the current operation temperature, and a current performance of the target component; and providing the maximum performance of the target component by using the human-machine interface.

9. The filter replacement reminder method according to claim 8, wherein the step of calculating the maximum performance of the target component by the inlet temperature, the rated temperature, the current operation temperature, and the current performance of the target component comprises: estimating the maximum performance of the target component based on the current performance of the target component by a ratio of a first difference between the rated temperature and the inlet temperature and a second difference between the current operation temperature and the inlet temperature.

10. A server device, comprising: a filter; a component temperature sensor; an ambient temperature sensor; a target component; a heat dissipation component; a human-machine interface; and a controller, coupled to the component temperature sensor, the ambient temperature sensor, the target component, the heat dissipation component, and the human-machine interface, wherein the controller is for: obtaining a current operation temperature of the target component by using the component temperature sensor; detecting a working state of the heat dissipation component; obtaining an ambient temperature by using the ambient temperature sensor; and providing a reminder for replacing the filter by using the human-machine interface by the current operation temperature, the working state of the heat dissipation component, and the ambient temperature, wherein the controller is for: providing the reminder to replace the filter by the working state of the heat dissipation component and the ambient temperature, in response to the current operation temperature being greater than or equal to a rated temperature of the target component.

11. The server device according to claim 10, wherein the controller is for: providing the reminder to replace the filter, in response to the working state of the heat dissipation component being normal and the ambient temperature being lower than an ambient temperature threshold.

12. The server device according to claim 10, wherein the heat dissipation component comprises a fan, and the working state of the heat dissipation component comprises a fan speed.

13. The server device according to claim 10, wherein the target component comprises a central processing unit, a hard disk, or a memory module.

14. The server device according to claim 10, wherein the controller is for: obtaining a previous operation temperature sensed at a previous sensing time, wherein the current operation temperature is sensed at a current sensing time; estimating a service life of the filter by a rated temperature of the target component based on a linear relationship of the current operation temperature and the previous operation temperature relative to the previous sensing time and the current sensing time; and providing a suggested service life of the filter by using the human-machine interface by the service life of the filter, wherein the previous operation temperature and the current operation temperature both correspond to a preset load state of the target component.

15. The server device according to claim 14, wherein the controller is for: estimating an another service life of the filter by the rated temperature of the target component based on an another linear relationship of the previous operation temperature and an another previous operation temperature relative to the previous sensing time and an another previous sensing time, wherein the suggested service life is the shorter of the service life and the another service life, and the previous operation

temperature and the another previous operation temperature both correspond to the preset load state of the target component.

16. The server device according to claim 14, further comprising an another component temperature sensor and the another target component, wherein the controller is coupled to the another component temperature sensor and the another target component, and is for: obtaining another current operation temperature and another previous operation temperature of the another target component by using the another component temperature sensor; and estimating another service life of the filter by an another rated temperature of the another target component, the another current operation temperature, and the another previous operation temperature, wherein the suggested service life is the shorter of the service life and the another service life.

17. The server device according to claim 10, further comprising an inlet temperature sensor, wherein the target component comprises a central processing unit, and the controller is for: obtaining an inlet temperature of the target component by using the inlet temperature sensor; estimating a maximum performance of the target component by the inlet temperature, a rated temperature, the current operation temperature, and a current performance of the target component; and providing the maximum performance of the target component by using the human-machine interface.

18. The server device according to claim 17, wherein the controller is for: estimating the maximum performance of the target component based on the current performance of the target component by a ratio of a first difference between the rated temperature and the inlet temperature and a second difference between the current operation temperature and the inlet temperature.